

CS 726: Problems on Bayesian Networks

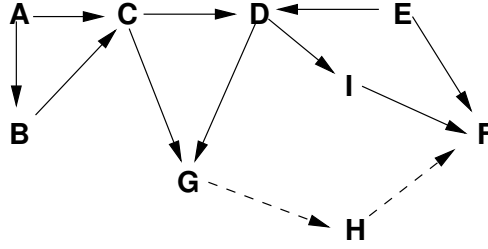
Write your answers in the space provided. You are expected to solve each question on your own. Do not try to search the answers from any external sources, like the web. You are allowed to discuss a few questions with your classmates provided you mention their names.

1. Draw a Bayesian network over five variables $x_1, \dots, x_5$ assuming the variable order x_1, x_2, x_3, x_4, x_5 . For this ordering, assume that the following set of local CIs hold in the distribution: $x_1 \perp\!\!\!\perp x_2$, $x_3 \perp\!\!\!\perp x_2|x_1$, $x_4 \perp\!\!\!\perp x_1, x_3|x_2$, $x_5 \perp\!\!\!\perp x_1, x_2|x_3, x_4$

2. Consider the Bayesian Network represented by the following graph:

- (a) Assume that the vertex H and the dotted edges $G \rightarrow H$ and $H \rightarrow F$ are absent. State which of these CIs are implied by a direct application of the Local-CI rule.

- i. $A \perp\!\!\!\perp G|F$
- ii. $A \perp\!\!\!\perp D$
- iii. $B \perp\!\!\!\perp G|C, D$
- iv. $F \perp\!\!\!\perp D|C, E$



- (b) Now add the vertex H along with the two dotted edges to the graph. Find the smallest set S ($A, E \notin S$) such that $A \perp\!\!\!\perp E \mid S$. Provide a proof if no such S exists.
- (c) In the above graph (without the dashed edges), if each variable is binary, how many parameters are required to represent the potentials of the distribution? How does that compare with number of parameters required by an explicit joint distribution?

3. Consider a distribution $P(x_1, x_2, x_3)$ as follows:

$$\begin{aligned} \Pr(x_1, x_2, x_3) &= 0.5 \text{ if } x_1 = x_2 = x_3 \\ &= 0 \text{ otherwise.} \end{aligned} \quad (1)$$

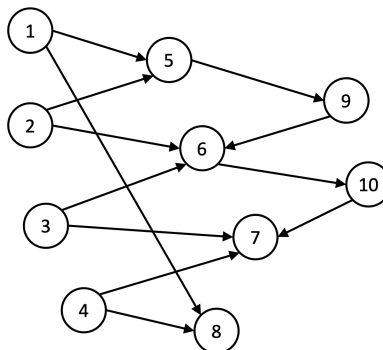
For this distribution $x_1 \perp\!\!\!\perp x_2 \mid x_3$ but say we change the distribution slightly to

$$\begin{aligned} \Pr(x_1, x_2, x_3) &= \frac{1-\epsilon}{2} \text{ if } x_1 = x_2 = x_3 \\ &= \frac{\epsilon}{6} \text{ otherwise.} \end{aligned} \quad (2)$$

For what values of ϵ is $x_1 \not\perp\!\!\!\perp x_2 \mid x_3$?

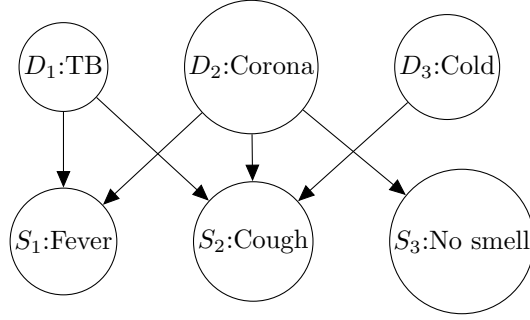
..3 We look for conditions under which $P(x_1 = 1 \mid x_2 = 1, x_3 = 1) = P(x_1 = 1 \mid x_2 = 0, x_3 = 1)$. That is, $\frac{1-\epsilon}{2} = \frac{\epsilon}{6}$. This holds only for $\epsilon = 0$ and for $\epsilon = \frac{3}{4}$. The latter corresponds to the case where the distribution is uniform over all 8 possible combinations. Students who point this point should get +2 extra credit.

4. For the BN below answer the following questions:



- (a) Provide the smallest subset of variables Z so that $x_5 \perp\!\!\!\perp x_4 \mid Z$..1 $Z = \phi$, they are already independent.

5. Consider a special QMR Bayesian network comprising of disease nodes $D_1, \dots, D_n$ and symptom nodes $S_1, \dots, S_k$. An example appears below.



Assume all variables are binary. Let probability of a disease D_i occurring be denoted as $P(D_i = 1) = \beta_i$ and α_{ij} denote the probability of a patient showing symptom S_j if he only has disease D_i . Also, assume the probability of a symptom not occurring conditioned on its parent nodes is $\Pr(S_j = 0 | Pa(S_j)) = \prod_{i \in Pa(S_j)} (1 - \alpha_{ij})^{D_i}$. For example, if someone has D_1, D_2 but not D_3 then the probability of his not showing symptom S_2 is $(1 - \alpha_{12})(1 - \alpha_{22})$

- (a) In a study over 1000 random Indians, 20 had Cold, 2 had TB, and 1 had Corona. Doctors have estimated that Corona causes each of these symptoms with probability 0.2, Flu causes cough with probability 0.7, and TB causes fever with probability 0.8, and cough with probability 0.3. Write exact numerical values for the following potentials as a table:

i. $P(D_1)$

..1

$D_1 =$	0	1
	0.998	0.002

ii. $P(S_3 | Pa(S_3))$

..2

$D_3 =$	0	1
$P(S_3) = 0$	1	0.3
$P(S_3) = 1$	0	0.7

iii. $P(S_1 | Pa(S_1))$

..2

$D_1, D_2 =$	0,0	0,1	1,0	1,1
$P(S_1) = 0$	1	0.2	0.8	0.16

- (b) What is the marginal probability $P(S_1 = 1)$ in the above example graphical model? Express only in terms of α_{ij} s and β_i s. ..2 $\sum_{D_1, D_2} \Pr(S_1 = 1 | D_1, D_2) P(D_1) P(D_2) = (1 - \alpha_{21})(1 - \alpha_{11})\beta_1\beta_2 + (1 - \alpha_{11})\beta_1(1 - \beta_2) + (1 - \alpha_{21})(1 - \beta_1)\beta_2 + (1 - \beta_1)(1 - \beta_2)$
- (c) If someone has Fever, No sense of smell, and No cough, what is the most likely disease that the person has assuming he has at most one disease? [Give a brief explanation for the answer.] ..2 TB: $2 \times 0.8 \times (1 - 0.3)$